Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	GenAI For Factory Insight			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication (), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		*Knowledge Services*	
	Medium Category	Small Category	Medium Category	Small Category
	Software	*Software Solution / System Integration*	*R&D / Engineering Services*	*Industrial Facility Survey / Design / Forecasting / Evaluation / Management Technologies*
Concept of Proposed Technology	Modern factory performance is no longer driven by a single machine metric, but by fast-changing trade-offs between throughput, delivery reliability, maintenance cost, energy use, quality, and work-in-progress ([7], [10], [13]). Today, many SMEs manage these trade-offs mainly through the experience of a few key people, experience that takes years to build and is difficult to scale or replace. When companies try to reduce this dependency with digital tooling, they often run into a practical barrier: the data exists, but it is scattered across systems and still requires scarce IT and data expertise to turn it into timely, actionable answers [8,9]. GenAI for factory insight addresses this gap by giving non-experts direct access to complex factory data, so managers, team-leads, engineers, and operators can generate management-relevant insights, validate hypotheses quickly, and act while issues are still relevant. GenAI for factory insight enables people to "talk to their factory data" by posing natural-language			

	questions and receiving clear, evidence-based answers grounded in actual production signals. It connects time-series data and events across machines, lines, MES/ERP, quality systems, and maintenance records to explain what happened, why it happened, and what is likely to happen next. The main function is to expose and reduce deviations between planned capacity, reported KPIs, and actual output, improving effectiveness, and lowering cost. The system explains KPI outcomes (e.g., OEE) by tracing losses to concrete drivers such as downtime, speed losses, micro-stops, and quality rejects. Using a Multi-Agent Generative System, specialized agents coordinate reliability, process optimization, quality control, and tactical planning decisions. Typical uses include daily performance review, real-time production support, scenario evaluation ("what if we change X?"), and cross-team coordination to resolve conflicts and improve overall line performance [11]. The expected impact is quicker problem solving, lower analysis workload, and faster operational improvements that translate into higher OEE, reduced scrap, and improved delivery performance.
Rationale for Joint R&D Project	Factories must continuously improve or risk losing competitiveness. Korean SMEs face the same pressure as Dutch manufacturers to make effective decisions faster and to act on issues while they still matter. However, developing GenAI capabilities for factory insight is technically and operationally complex and therefore risky to pursue alone. First, high-quality data from machines and shopfloor/business systems must be available. In practice, data quality and integration are the main bottlenecks. TNO has extensive experience integrating heterogeneous equipment into (generic) IoT platforms [3] and applying Industry 4.0 concepts and standards ([1], [4], [5], [15]), creating reusable digital foundations that can also accelerate other initiatives beyond this project. Second, AI specialists with experience in manufacturing are needed to select and implement the right kind of algorithms. TNO has a proven track record in applied manufacturing innovation, translating industrial challenges into working solutions in projects ([2], [6]), and advancing factory autonomy in programs such as [13]. Third, GenAI evolves rapidly and must be engineered carefully to avoid failure modes such as hallucinations that can mislead decision makers [12]. In [14] we developed and tested an LLM-based control loop to support operators in a one-piece-flow production setting. Finally, successful adoption requires alignment with long-term strategy, business value, and change management ([7], [10]). With TNO, the Korean SME gains a partner that accelerates data readiness, builds a working prototype fast, and makes sure the solution is safe, reliable, and useful for daily operations. From TNO's perspective, robust and generalizable methods for "talking to factory data" require validation in real industrial environments where data issues, system interfaces, and operational constraints can be observed and solutions tested in practice. SMEs are an ideal partner because their scale enables rapid iteration, direct access to domain experts, and practical validation cycles, while still representing the real-world complexity of manufacturing operations. Collaboration is necessary because neither party can achieve the objective alone: TNO can de-risk and structure development using proven Industry 4.0 methods and state-of-the-art GenAI techniques, while the SME provides operational context, data access, and continuous feedback that determines whether the solution works under real conditions. Joint execution ensures the technical choices remain aligned to practical constraints and deliver measurable impact. Ultimately, the aim is to turn state of the art research into a solution that the SME can use in daily operations and expand over time, so it supports both today's improvements and future innovation.
Key Words	1. Autonomy 2. Decision-Making 3. Gen-AI 4. Industry 4.0 5. Manufacturing KPIs

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3-4)	End of the Project	Level (6)
R&D Objectives	**GenAI for Factory Insight:** • **SME specific Unified Namespace (UNS):** structure and contextualize the SME's factory data, preparing it for reliable GenAI querying and reasoning across their systems. • **Multi-modal data ingestion and processing pipeline**: Collect, clean, align, and store time-series and related data in a database suited for GenAI use. • **Time-series "memory" generation**: Convert high-frequency sensor and event streams into compact, queryable memories for GenAI reasoning. • **Integrate multi-level reasoning methods**: Improve answering quality by combining descriptive, diagnostic, and predictive reasoning over factory context. • **Human-in-the-loop interaction and control**: Allow users to review evidence, give feedback, approve actions, and steer the system safely. • **Web-based chat interface**: intuitive human interaction through natural-language questions. • **Multi-modal outputs for decision support**: Depending on this specific SMEs needs, dashboards, graphs, planning suggestions, documentation, and example tasks tailored to different roles.			
R&D Target Specifications	***Connected information sources (coverage)*** • Metric: Number of distinct, production-relevant sources integrated and queryable through the GenAI layer (e.g., MES, ERP, SCADA/PLC/historian, machine controllers, quality system, CMMS, manuals/SOPs). • Target: 4+ connected, of which 2+ are shopfloor critical • Acceptance check: Data from each source is searchable and can be referenced in answers (with source attribution). ***Confirmed hypotheses rate (decision usefulness)*** • Metric: Percentage of hypotheses where the tool provides sufficient evidence to reach a clear conclusion (confirmed or disproven), validated by an SME domain expert. • Target: ≥ 50% of the representative hypothesis set reaches a clear conclusion, with documented evidence links. ***New insights generated (novelty)*** • Metric: Count of insights identified by the system that were previously unknown to the SME. • Target: ≥ 5 validated new insights per month during the evaluation period ***Operationalizable insights (actionability)*** • Metric: Percentage of insights that translate into an executable action (process tweak, parameter change, maintenance intervention, planning adjustment) with an identified owner and expected effect. • Target: ≥ 25% of validated insights are operationalizable ***Questions answered (service level and correctness)*** • Metric: Number of natural-language questions answered end-to-end with traceable evidence, plus answer quality. • Target: ≥ 200 questions answered during the pilot; ≥75% rated "correct and useful" by users.			

R&D Plan

Research approach

The project combines Industry 4.0 principles (UNS, IoT, ISA-95, ISO-22400) with GenAI interaction patterns (tool calling, structured outputs, human-in-the-loop). It follows an iterative approach: (1) formalize the SME's processes, roles, and KPIs, (2) structure and connect data early via an SME-specific Unified Namespace (UNS) and IoT connectivity, (3) run a state-of-the-art study to select and prototype the most suitable algorithms for converting industrial time-series into GenAI-ready "memories," (4) enable user interaction through a working chat-to-data prototype using the latest GenAI principles, and (5) progressively extend the prototype with document grounding (RAG), multi-level reasoning, and human-in-the-loop controls. Process modelling and KPI definitions will be aligned with international standards and best practices. The project deliberately front-loads transfer of established Industry 4.0/UNS know-how to accelerate data readiness and reduce integration risk.

Throughout the project, regular interactive working sessions with the SME guide algorithm choices, interface design, and use-case prioritization, ensuring each iteration delivers a usable prototype that is continuously improved and validated.

Phases, key tasks, methods, milestones, timeline

The project execution is split into six distinct phases, each of which can take between 3 and 6 months, depending on the concrete use case at hand, and the development results. In the situation when phase 6 is reached prior to the end of the project duration, an extra phase will be introduced, focused on the introduction of more complex functionalities to the system or expansion of the processes.

Phase 1: Baseline and problem framing

Tasks

- Inventory of factory processes, roles, and KPIs (including KPI definitions and calculation logic).
- Inventory of software packages and data flows.
- Inventory of machines and their software interfaces
- Define the priority "analysis questions" and hypothesis classes to benchmark (downtime causes, micro-stops, speed losses, defect spikes).

Methods

- BPMN, ISA-95 mapping, ISO-22400 KPI dictionary and traceability.

Milestones

- M1: Process maps + KPI dictionary.
- M2: Software/data-flow gap analysis + benchmark question set.

Phase 2: UNS foundation and data readiness

Tasks

- Design the SME-specific UNS, following UNS design recommendations and i4.0 principles
- Connect the identified production-relevant sources (see phase 1) to the UNS.
- Apply the TNO UNS MCP server to the UNS and expose tools that can query the raw signals
- Collect manuals/SOPs/schematics for later use.

Methods

- UNS patterns, IoT integration, AAS-inspired semantic modelling where beneficial.

Milestones

- M3: UNS v1 with initial connected sources and validated tags/events.

Phase 3: Build prototype for time-series -> "memories"

Establish the algorithmic foundation for turning time-series into GenAI-friendly representations that enable accurate explanations, faster hypothesis confirmation, and scalable reasoning.

Tasks

- State-of-the-art review of time-series representation and summarization approaches for LLM consumption.
- Create a benchmark dataset from the SME environment (representative lines/assets)
- Build prototype pipeline to convert selected signals
- Evaluate prototype with benchmark dataset

Methods

- Time-series segmentation and feature extraction; symbolic/time-window summarization; anomaly detection; embedding-based retrieval; evaluation protocol with expert validation.

Milestones

- M4: Prototype "memory pipeline" to produces queryable memory objects linked to raw data for traceability.

Phase 4: Build Human-in-the-Loop interfaces

Tasks

- Build chatbot interface to "talk to their factory data"
- Extend chatbot interface with human-in-the-loop controls (review, feedback, approvals) to improve GenAI recommendations and measure R&D target specifications
- Prompt engineering and response structuring (sources, assumptions, uncertainty, recommended checks).

Methods

- Tool calling design, function schemas, guardrails; HMI design; structured outputs.

Milestones

- M5: Prototype v1: chat-to-UNS with memory-aware answers and evidence linking.

Phase 5: Extend with RAG-based documents and reasoning engine

Tasks

- Extend solution with RAG over manuals/SOPs/schematics for contextual explanations and "how-to" guidance.
- Add a reasoning engine to extend the reasoning capabilities of LLMs.

Methods

- RAG pipeline tuning; multi-step reasoning orchestration.

Milestones

- M6: Prototype v2: integrated RAG + reasoning engine.

Phase 6: Pilot validation

Tasks

- Pilot in operations; measure against quantitative target specs (sources connected, time-to-confirm hypotheses, new/actionable insights, question throughput, user-rated usefulness).
- Technology transfer to SME

Methods

- Field validation; KPI impact measurement; benchmarking.

	Milestones • M7: Final architecture + report against targets + rollout roadmap (cloud/on-prem options) and scale-up plan.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• **Apply Industry 4.0 principles in the SME context:** translate ISA-95 and ISO-22400 concepts into a practical, SME-specific data and KPI framework, and guide the setup of an SME-specific UNS. • **Provide UNS expertise and assets**: deliver and tailor TNO UNS guidelines, including naming conventions, data semantics, and integration patterns to ensure consistent and scalable data structuring. • **Support IoT connectivity and data acquisition**: advise on IoT architectures and interfaces to extract and contextualize data from machines and shopfloor systems, including practical integration options and constraints. • **Support process formalization where needed**: consult the SME on documenting processes and responsibilities (e.g., BPMN-style descriptions) to link data, KPIs, and operational workflows. • **State-of-the-art research and selection**: perform and document the SoTA analysis for time-series-to-GenAI representations, define evaluation criteria, and recommend the most suitable algorithms for the SME use cases. • **Develop the GenAI proof-of-concept for factory insight**: implement a working POC that enables talking to the SME's factory data. This work will be an extension of existing TNO GenAI tooling for UNS.
Expected R&R of Korean SME Partner	• **Define business goals and KPIs**: clarify the SME's priority business KPIs and operational targets (e.g., productivity, quality, downtime), including definitions, calculation rules, and accepted "ground truth." • **Provide process knowledge and formalization**: share detailed information about shopfloor and support processes and collaborate on creating formalized process descriptions (including roles, handovers, and key decision points). • **Enable data access and integration**: provide access to relevant factory data via APIs and interfaces (MES/ERP, machine data, quality, maintenance), and support data governance, permissions, and connectivity approvals. • **Extend machine/data connectivity where needed**: contribute technical resources to connect machines or data sources that are not yet reachable via IoT, including troubleshooting and onboarding of legacy equipment. • **Co-develop integration at SME**: allocate technical staff to co-develop integration components, configuration, and deployment elements, using the SME's IT infrastructure and engineering practices. • **Provide test cases and insight hypotheses**: supply prioritized "insight questions," hypotheses, and known operational issues to drive development, benchmarking, and algorithm evaluation. • **Pilot usage and validation**: use the developed tooling in real operational routines (e.g., daily performance reviews, incident analysis), evaluate usefulness and correctness, and provide structured feedback for iterative improvement.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project will deliver a working GenAI application for talking to factory data. A GenAI assistant that can answer natural-language questions with traceable evidence from shopfloor and business systems. A key technical outcome is a validated time-series-to-"memory" method, selected through state-of-the-art research, enabling fast and reliable explanations of downtime, speed losses, micro-stops, and quality issues. In daily use, teams will confirm or reject performance hypotheses much faster and generate more actionable insights without relying on scarce IT and data experts. The expected business impact is quicker problem solving, lower analysis workload, and faster operational improvements, which in turn will result in higher OEE, reduced scrap, and improved delivery performance. **Commercialization potential**: The GenAI assistant can be productized and offered to other factories as a self-service factory insight layer and decision-support layer. Built on a standardized UNS approach, it can be deployed repeatably across lines and sites and integrated with common shopfloor and business systems, enabling a scalable go-to-market as a software product or solution offering. Differentiation comes from reducing analysis effort for daily operations while improving decision speed and consistency through traceable insights. **Economic potential**: Factory optimization has shifted from a single-focus performance topic to a multi-variable optimization problem in which throughput interacts with delivery reliability, maintenance cost, energy use, and work-in-progress levels. In this setting, managers, team-leads, and process engineers cannot realistically keep all constraints, trade-offs, and fast-changing signals in their heads. They need decision support that can quickly surface the right insights, explain key drivers, and recommend practical actions while there is still time to influence outcomes. By shortening the cycle from question to validated insight to action, the proposed GenAI assistant helps reduce avoidable losses and improves overall operational and financial performance.

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